

RTL Design Sherpa

DDR3 / LPDDR3 Family Controller Hardware Architecture Specification 0.9

October 3, 2026

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1 Scoria Has Index

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2 Document Information

Table 0.1: Document information

Field	Value
Title	scoria DDR3/LPDDR3 Family Controller — Hardware Architecture Specification
Version	0.12
Date	2026-10-20
Status	Advanced-modes edition. v0.8 reconciled this book with the landed RTL; v0.9 specifies TASK-001's bounded advanced modes (elastic refresh, TCR, ZQCS placement) in Ch 3.2/3.4 per that RTL; v0.10 restates the design-point clock at 75 MHz per the owner (2026-10-10: "100 MHz was never the target"), closing scoria BUG-003 — revision history below
Scope	Controller architecture to the DFI v3.1 boundary, for DDR3 and LPDDR3
Not in scope	The PHY; board bring-up; the DDR4/LPDDR4 features DFI v3.1 also carries

2.1 Related documents

Table 0.2: Related documents

Document	Where	Relationship
scoria design requirements	`../design-requirements.md`	**binding foundation** — the spec deltas and settled decisions
pumice HAS	`../../pumice-ddr2-lpddr2/docs/pumice_has/`	the architecture scoria is derived from
pumice design requirements	`../../pumice-ddr2-lpddr2/docs/design-requirements.md`	its coding guidelines and enforcement rules apply here unchanged
JESD79-3F	outside the repo, cold storage	DDR3 device standard
JESD209-3C	outside the repo, cold storage	LPDDR3 device standard
DFI v3.1	`dfi-specs/`	the PHY boundary

2.2 Terminology

INHERITED / MODIFIED / NEW The marking every block in this document carries, relative to pumice. *Inherited* means the pumice block is used as-is, and its correctness argument transfers with it. *Modified* means a named, bounded change. *New* means no pumice counterpart exists.

nCK A DRAM clock cycle, the unit JESD79-3F states most command-spacing minimums in.

Prime DQ bit In write leveling, the DQ bit (or bits) on which the DRAM returns the leveling result.

Maintenance traffic Commands the controller must issue that no host requested — refresh and ZQ calibration. It competes with demand traffic for the command bus, which makes it a scheduling problem and not only a sequencing one.

2.3 Revision History

Table 0.3: Revision history

Version	Date	Change
0.10	2026-10-10	Design-point clock restated at **75 MHz** per the owner (2026-10-10: “100 MHz was never the target”). Ch 2.4 Table 2.4 (`System clock` row and the `DRAM clock` derivation — the PHY clock comes from the 200 MHz board oscillator’s PLL, not `4 x sys`) and the “still open” timing-closure bullet updated; scoria BUG-003 closed on the restatement (its OOC measurement, -2.022 ns at 10 ns, is ~+1.3 ns of slack at 13.33 ns). DRAM side unchanged: DDR3-800, 400 MHz CK, 3200 MB/s peak.
0.9	2026-10-03	TASK-001’s bounded advanced-modes tranche specified from the landed RTL: Mode A (demand-aware elastic refresh) and Mode B (TCR) in Ch 3.4, Mode C (ZQCS placement policy) in Ch 3.2, with the mode-select CSRs and `REF_STATS` telemetry in Ch 5. `refresh_ctrl` marking INHERITED → MODIFIED; the Ch 2 block diagram

Version	Date	Change
		(figure re-rendered), module hierarchy, and Ch 3.1 deltas table updated to match (inherited FUB count 21 → 20). The refresh/ZQ portion of the TASK-008 reconciliation is thereby done; the remainder — the other inherited blocks, `tWLMRD`'s policy value, and whole-map CSR finalization — stays open in Ch 6.
0.8	2026-10-03	Reconciled with the landed RTL and the first board work: verification status brought current (221 tests, 9 formal blocks, top tier AXI-in/DFI-out); `dfi_reset_n` recorded as presented under its DFI name and the two v3.1-only data-phase chip selects recorded as driven, constant `'0` at `NUM_RANKS=1` (scoria TASK-005); the board build flow and the BUG-003 100 MHz out-of-context timing result recorded. What remains for a 1.0 is restated in Chapter 6 and filed as a task.
0.7	2026-09-30	Ch 3.2: DDR3's init sequence needs no

Version	Date	Change
		precharge-all, no double refresh, no OCD pair and no second MR0 load (the DLL-reset bit is self-clearing) – the FSM drops seven of pumice's states and adds three. Found while implementing.
0.6	2026-09-30	Corrected during implementation: `dfi_signal_pack` is INHERITED, not MODIFIED – it packs only signals DFI v3.1 leaves unchanged, and v3.1's new channels are driven at the plane above. Inherited FUBs 20 -> 21, modified 4 -> 3.
0.5	2026-09-30	Target design point named (Sean: "assume k7ddrphy on genesys2 if it helps"): Genesys 2, K7DDRPHY, 2 x MT41J256M16 on a 32-bit bus, DDR3-800, 3200 MB/s theoretical peak. Fixes DFI_RATE=4, NUM_BANKS=8, row 15, col 10, DQ 32, and gives the DDR3-800 JEDEC timing set.
0.4	2026-09-30	Q1 and Q4 answered rather than deferred: s7ddrphy is the PHY for

Version	Date	Change
		every board here, it implements no DFI low-power interface, and it imposes no leveling timeout. `powerdown_ctrl` corrected to INHERITED in mechanism – power-down is CKE and SRE/SRX, not the DFI low-power channel. No open questions remain.
0.3	2026-09-30	Q2 and D2 re-answered from a generated DDR3 LiteDRAM core instead of by analogy (Sean: “you might need to generate a ddr3 version of litedram”). ZQCS confirmed as request/grant sharing the refresher FSM; write leveling confirmed as CSR-only with no state machine.
0.2	2026-09-29	Q1-Q5 resolved or deferred with conditions. `refresh_ctrl` corrected from MODIFIED to INHERITED: pumice already implements `REFpb` and the device owns the sequence, so Q5 was malformed and is struck.
0.1	2026-09-29	First edition. Written from the delta analysis, with

Version	Date	Change
		decisions D1-D3 settled. No RTL exists; every block is specified, not described.
0.11	2026-10-20	Block diagrams redrawn to the repo element key (v3): class fills, double-bordered registered blocks, green return paths and a generated key under each figure. No technical content changed.
0.12	2026-10-20	Section (h2) headings now open a new page, the pipeline default. No technical content changed.

2.4 How to read this book

Chapter 1 fixes purpose and vocabulary. Chapter 2 is the shape of the controller, with one figure in which every block is colour-marked against pumice — read that figure and you know the size of this project. Chapter 3 is the only chapter with substantial new engineering: the four blocks DDR3 and LPDDR3 actually change. Chapter 4 is the DFI v3.1 boundary and what moved from v2.1.1. Chapter 5 is the package and parameters. Chapter 6 is how it is verified, and what separates this edition from a 1.0.

A caution this document outgrew. v0.1 was written before the RTL, and a specification written first can be *unbuildable* — that was the risk it carried. The RTL now exists, and this edition has been reconciled against it; where implementation corrected the specification, the correction is in the revision history above. What remains for a 1.0 is named in Chapter 6 and filed as a task. The evidentiary rule is unchanged: where this edition states a structure, it is inherited from a controller that is built and measured, or built here and verified; where it states a timing, it is cited.

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3 Purpose and Scope

3.1 Purpose

scoria is a unified, parameterized memory controller for DDR3 SDRAM and LPDDR3 SDRAM, presenting an AXI4 slave to the host and a DFI v3.1 master to the PHY. It is the second member of a family: pumice covers DDR2/LPDDR2 and is built, measured and running on hardware; a DDR4/LPDDR4 controller is planned as the third.

This document specifies scoria's architecture. It was written to be *locked* before RTL began, so that the implementation had a fixed target; the RTL now exists, and each edition since has reconciled the two. The PRD — which explicitly waited on a locked HAS — is authored against this document.

3.2 Why a delta specification

scoria is not a fresh design. The controller architecture it uses is pumice's, and that architecture has been through board bring-up, a read-path rebuild that took reads from 291.7 to 571.3 MB/s, and a formal campaign that found two real bugs. Re-deriving it would discard that evidence.

So this specification is organised as a delta. Every block is marked **INHERITED**, **MODIFIED** or **NEW**, and the marking carries an obligation:

- **INHERITED** means the pumice block is used unchanged, and its verification evidence transfers. If a block is marked inherited and then changed during implementation, the marking is wrong and this document must be corrected — not the marking quietly dropped.
- **MODIFIED** means a bounded change with a named cause, always a clause of JESD79-3F, JESD209-3C or DFI v3.1.
- **NEW** means no pumice counterpart. There are two, and both are small.

3.3 What is in scope

The controller between the AXI4 slave port and the DFI v3.1 master port: the front end, the scheduler, bank and global timing enforcement, refresh, mode registers and initialization, ZQ calibration, the write-leveling interface, the data paths, and the CSR block.

3.4 What is out of scope

The PHY. DFI is the boundary, as it is for pumice, and for the same reason: IOB serdes, bitslip and delay-line calibration are FPGA- or process-specific and do not belong in a portable controller. This is a deliberate architectural commitment, not an omission.

The write-leveling search. DDR3 adds write leveling, and scoria provides the interface to perform it — the DFI handshake, the MR1 path, the timing windows and the telemetry. The search algorithm itself runs in firmware. This was settled as decision D2 and is argued in Chapter 3.

The DDR4/LPDDR4 half of DFI v3.1. Targeting v3.1 brings signals scoria has no use for: `dfi_act_n`, bank groups, chip ID, CA parity and `dfi_alert_n`, Data Bus Inversion, and CA training. They are named here so their absence is a decision on the record rather than an oversight.

Board bring-up. Verification is against a DFI bus functional model in simulation, and that remains the verification boundary. The board target is no longer deferred — Genesys 2, K7DDRPHY, 2 x MT41J256M16 (Chapter 2.4) — a board build flow exists, and the first out-of-context synthesis has been run (scoria BUG-003: WNS -2.022 ns against a 10 ns constraint, which closes with ~+1.3 ns of slack against the restated 75 MHz design point; the owner restated the clock 2026-10-10 — 100 MHz was never the target — and the bug closed on that restatement). Bring-up itself remains outside this document's scope.

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4 Document Conventions

4.1 Block marking

Every block carries exactly one marking, and the block diagram colours match:

Table 1.1: Block markings

Marking	Colour in Figure 2.1	Meaning
INHERITED	green	pumice’s block, used as-is; its verification evidence transfers
MODIFIED	amber	a bounded change with a cited cause
NEW	red	no pumice counterpart

4.2 Signal and parameter naming

Inherited from the pumice design requirements and not restated here: module naming, `UPPER_CASE` parameters, `r_`/`w_` internal prefixes, and the per-area clock and reset names. Two points bear repeating because they are load-bearing:

- **Clock and reset names are per area.** In the component tree the controller uses `aclk`/`aresetn`. An `.i_rst_n(...)` connection against an `rtl/common` module will not elaborate, because those use `clk`/`rst_n`.
- **No assertions in the RTL.** Properties live in external `formal/` blocks. This is a tool-compatibility rule, not a style preference.

4.3 Timing values

Every enforced timing is a runtime CSR, not a parameter. This is inherited from pumice and was learned there: a timing compiled in cannot be swept, and a controller whose timings cannot be swept cannot be characterized.

Where this document states a numeric timing it is cited to a clause, and the unit is given. `nCK` means DRAM clock cycles. Values that vary by speed bin are named but not tabulated here — they belong to the CSR derivation.

Note: JESD79-3F states some minimums without a maximum, most notably `tWLMRD`, whose maximum it declares controller-dependent. Where the spec declines to bound something, this document says so rather than inventing a bound.

4.4 Citations

A claim traceable to a standard is cited as JESD79-3F (DDR3), JESD209-3C (LPDDR3), or DFI v3.1 with a section number. Those PDFs are **not in the repository** and must not be added to it — they are not redistributable. They live in the operator's cold storage; `dfi-specs/ddr3/index.md` records where and why.

A claim inherited from pumice cites the pumice HAS or its design requirements, both of which are in the repository.

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5 Definitions and Acronyms

5.1 Interfaces

AXI4 The host-side slave interface. Inherited from pumice unchanged, including the width-gearing converters that let the host bus width differ from the DRAM's.

DFI DDR PHY Interface. scoria targets **v3.1** (decision D1). The boundary between controller and PHY, and the same boundary the DV repo's bus functional model drives in simulation.

APB The CSR access interface, through a PeakRDL-generated register block.

5.2 DRAM commands this document refers to

Table 1.2: Commands, and which are new to this family member

Command	Present in DDR2?	Note
ACT, RD, WR, PRE, REF	yes	inherited handling
PREA	as PRE with A10	DDR3 gives precharge-all its own encoding
MRS	yes	but DDR3 has MR0-MR3 where DDR2 had MR0-MR2 plus EMRS3
ZQCL / ZQCS	**no**	ZQ calibration, long and short. New in DDR3
RESET	**no**	DDR3 adds a hardware RESET# pin
SRE / SRX	yes	self-refresh entry and exit
REFpb	LPDDR2 has it	LPDDR3 keeps per-bank refresh

5.3 Timing parameters new to scoria

Table 1.3: New timing parameters

Parameter	Governs
`tZQinit`	ZQ calibration during initialization
`tZQoper`	a long ZQ calibration outside initialization
`tZQCS`	the short, periodic ZQ calibration
`tWLMRD`	delay from entering write-leveling mode to the first DQS pulse
`tWLDQSEN`	delay before the controller may drive DQS, by which point the DRAM has applied ODT
`tWLO`	write-leveling output delay: when the DRAM returns the result
`tWLOE`	DQ output uncertainty, allowing mismatch across DQ bits

All seven are runtime CSRs. `tZQinit`, `tZQoper` and `tZQCS` also gate command issue alongside `tMRD`, `tMOD` and `tRFC` (JESD79-3F).

5.4 Terms used with a specific meaning here

Maintenance traffic Commands no host requested: refresh and ZQ calibration. Because it competes for the command bus, it is the scheduler's concern and not only the sequencer's.

Demand traffic Reads and writes originating from the AXI4 port.

Prime DQ bit The DQ bit (or bits) carrying the write-leveling result back from the DRAM.

Firmware-driven Performed by software over the CSR interface rather than by a hardware state machine. Both PHY leveling on pumice's board and scoria's write-leveling search are firmware-driven, deliberately.

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6 Scope and Goals

6.1 What scoria is

A memory controller presenting an AXI4 slave to the host and a DFI v3.1 master to the PHY, supporting DDR3 SDRAM and LPDDR3 SDRAM from one parameterized design.

6.2 Goals

1. **Reuse the pumice architecture, and its evidence with it.** The front end, scheduler, timing enforcement and data paths are inherited. Reuse here is not laziness; those blocks carry board measurements and formal proofs that a rewrite would discard.
2. **Change only what the standards force.** Four blocks change and two are new. Chapter 3 is that list, and it is short on purpose.
3. **Keep the DFI boundary clean.** No PHY responsibilities migrate into the controller — including the write-leveling *search*, which DDR3 makes tempting and decision D2 rules out.
4. **Every enforced timing a runtime CSR.** Inherited from pumice, and the precondition for characterization.
5. **Verifiable before hardware.** The DFI bus functional model is the target, so the controller can be wrong in simulation where it is cheap.

6.3 Non-goals

Performance parity with a vendor controller at DDR3 speeds. scoria's design point is architectural clarity and measurability. pumice reaches roughly 95% of theoretical peak in both directions on its board, and the same structure should carry, but no throughput target is set in this edition.

The DDR4 feature set. DFI v3.1 carries it; scoria does not implement it.

Multi-rank beyond pumice's parameterization. `NUM_RANKS` is inherited. As with pumice, the practical limit is the board, not the controller.

6.4 What makes DDR3/LPDDR3 easier than it looks

Three things reduce the work substantially, and it is worth stating them so the project is not over-scoped:

- **LPDDR3 is not a protocol break.** JESD209-3C keeps the 10-bit double-data-rate CA bus that LPDDR2 uses, so the LPDDR3 side is timings and mode registers rather than a new command encoding.
- **pumice was written expecting this.** Its row field is already padded to 18 bits with the comment “DDR3 forward-compat”, and `memtype_e` is an enum rather than a boolean — though a one-bit one, which is why decision D3 gives scoria its own package.
- **DFI v3.1 does not disturb the datapath.** Its multi-phase signalling generalises pumice’s explicit `_p0..p3` to `_pN`, and the 1:4 frequency ratio is still defined. `DFI_RATE = 4` carries over untouched.

6.5 What is genuinely harder

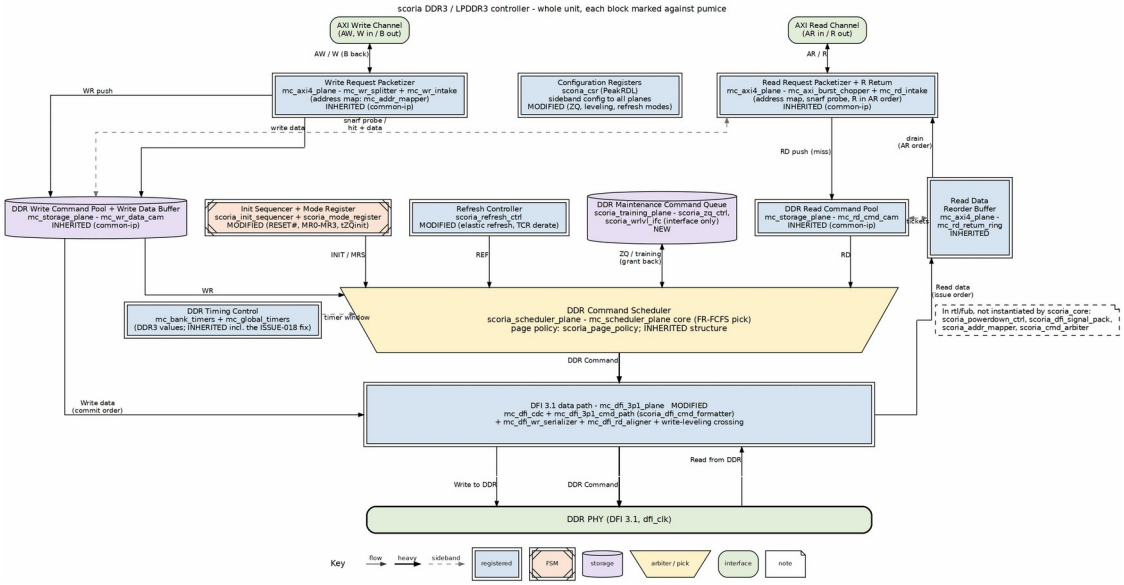
- **Write leveling.** New to this family member, and it spans the controller, the DRAM's MR1 and the PHY. The interface is specified here; the search is not.
- **ZQ calibration as maintenance traffic.** Periodic `ZQCS` has to be interleaved with demand traffic, which makes it a scheduling problem.
- **A four-register mode-register set.** MR0-MR3 replaces MR0-MR2 plus EMRS3, and the programming order is JEDEC's. pumice learned that the hard way: its init order was corrected to JEDEC's after an EMRS3-first sequence, which was benign but wrong.

7 Block Diagram

7.1 The whole controller, marked against pumice

Read this figure first. The last line of each block's label tells you the size of this project: INHERITED blocks are pumice's (the common-ip planes), used as-is; MODIFIED blocks change in named, bounded ways; NEW blocks are new, and there are two. The dashed box lists the scoria FUBs that are in `rtl/fub` but that `scoria_core` does not instantiate.

7.1.1 Figure 2.1: scoria block diagram, with inheritance marking



scoria DDR3/LPDDR3: each block marked INHERITED, MODIFIED or NEW against pumice

Source: 01_block_diagram.dot

7.2 Reading the figure

The front end is entirely INHERITED. The AXI4 interface — 1:1 read and write intakes, the write-data CAM, the read-reorder CAM, in-order commit, and the FSM-free split/aggregate — is untouched by the move to DDR3. Nothing about DDR3 or LPDDR3 reaches the host side.

The scheduler is INHERITED in structure, and its timing *values* change. The FR-FCFS arbiter, the per-(rank,bank) bank timers and the global timers are inherited. What changes is the numbers they enforce, and those are runtime CSRs rather than structure.

Important: one inherited detail must be inherited *including its fix*. pumice's `global_timers` published readiness flags one cycle late — registered status sampled the current state instead of the next — so the flags alone permitted violations of tCCD and tRTW. That was pumice ISSUE-018, and the fix was to compute one next-state function and feed both the counter and its status flop from it. scoria inherits the fixed form. A fresh implementation of the same block from the same description would reintroduce the bug.

The MODIFIED blocks are the three FUBs the standards change: the init sequencer, the mode-register block, and `dfi_cmd_formatter` for the new command encodings — plus the CSR block, which grows to hold the new timings. v0.9 adds a fourth FUB to the MODIFIED set: `refresh_ctrl`, for TASK-001's Modes A/B (Chapter 3.4) — a CSR-selected scheduling-policy plane, not a mechanism change.

Note: power-down was marked INHERITED, and MODIFIED in earlier editions of this book: it works through CKE and SRE/SRX, and the DFI low-power channel that v0.2 built a requirement around is not implemented by this PHY family at all (Chapter 3.1). Refresh was corrected to INHERITED the same way — pumice already implements REFpb and the device owns the sequence (Chapter 3.4) — and v0.9 turns it MODIFIED again, for a different reason: TASK-001's Modes A/B add a CSR-selected scheduling-policy plane (elastic refresh, TCR) on top of the inherited mechanism. Both corrections came from reading an artifact instead of reasoning about one.

`dfi_signal_pack` went the same way during implementation: marked MODIFIED for the "v3.1 signal set", it turned out to pack only signals v3.1 leaves alone.

The NEW blocks are the two new ones, and both are deliberately small: a ZQ controller that issues periodic ZQCS as maintenance traffic, and a write-leveling interface that contains no search loop.

7.3 What the figure deliberately omits

The PHY, because DFI is the boundary. A tap-search state machine, because decision D2 puts the search in firmware. And DFI v3.1's DDR4/LPDDR4 signals, because scoria does not implement them. This section says so, so that their absence reads as a decision rather than an omission.

RTL check (2026-10-13). `scoria_powerdown_ctrl`, `scoria_dfi_signal_pack`, `scoria_addr_mapper` and `scoria_cmd_arbiter` exist in `rtl/fub` but nothing in `scoria_core` instantiates them: the scheduler pick is the common `mc_cmd_arbiter` inside `mc_scheduler_plane`, the address map is the common `mc_addr_mapper` inside the intakes, and the DFI pins come from `mc_dfi_3p1_plane`. The figure lists them in its dashed box rather than drawing them on a flow.

8 Module Hierarchy

8.1 Planned hierarchy

scoria mirrors pumice's three-tier shape — FUBs under macros under a top — with the `scoria_` prefix replacing `pumice_`. The table below is the planned module list with its marking; it is a specification, and the RTL is expected to match it or this document to be corrected.

8.1.1 Top and macro tier

Table 2.1: Top and macro tier

scoria module	pumice counterpart	Marking
`scoria_top`	`pumice_top`	INHERITED (structure)
`scoria_top_geared`	`pumice_top_geared`	INHERITED — the host/DRAM width-gearing wrapper
`scoria_core`	`pumice_core`	INHERITED
`scoria_axi4_plane`	`pumice_axi4_ifc`	INHERITED
`scoria_scheduler_plane`	`pumice_mem_cmd_scheduler`	MODIFIED — must admit ZQ maintenance demand
`scoria_dfi_plane`	`pumice_dfi_plane`	MODIFIED — v3.1 control surface

8.1.2 FUB tier, inherited unchanged

`addr_mapper`, `bank_timer`, `scoria_bank_timers`, `global_timers` (in its fixed form — see Chapter 2.2), `scoria_cmd_arbiter`, `scoria_page_policy`, `scoria_rd_cmd_cam`, `scoria_wr_data_cam`, `scoria_rd_intake`, `scoria_wr_intake`, `scoria_wr_splitter`, `scoria_rd_return_ring`, `scoria_axi_burst_chopper`, `scoria_dfi_cdc`, `scoria_dfi_cmd_path`, `scoria_dfi_rd_aligner`, `scoria_dfi_wr_serializer`, `scoria_cmd_history_checker`, and **`powerdown_ctrl`** – power-down is CKE plus SRE/SRX on this PHY family, so only the new command encodings change, and **`dfi_signal_pack`** – a pure registered pipeline stage whose packed signals v3.1 does not touch.

That is 20 of the 24 FUBs carried over with no functional change; the count was 21 through v0.8, and `refresh_ctrl` is the one that moved — into the modified table below, for TASK-001's Modes A/B (Ch 3.4). The command history checker is worth naming explicitly: it is the verification-side block that independently re-derives JEDEC spacing from the issued command stream, and it must grow DDR3's parameters, but its mechanism is inherited.

8.1.3 FUB tier, modified

Table 2.2: Modified FUBs

Module	Why it changes	Detail
`init_sequencer`	DDR3 adds a `RESET#` pin and a four-register MR set	Ch 3.2
`mode_register`	MR0-MR3 replaces MR0-MR2 plus EMRS3	Ch 3.2
`dfi_cmd_formatter`	new command encodings: `ZQCL`, `ZQCS`, `PREA`	Ch 3.2
`refresh_ctrl`	TASK-001 Modes A/B — elastic pull-in/postpone streaks and the TCR tREFI derate behind CSRs. The `REFpb` rotor and the JEDEC ±8 credit accumulator are inherited unchanged	Ch 3.4

8.1.4 FUB tier, new

Table 2.3: New FUBs

Module	Purpose
`scoria_zq_ctrl`	issues `ZQCS` periodically as maintenance traffic; interval a runtime CSR; CSR-selectable placement policy (TASK-001 Mode C). `ZQCL` is init-only and belongs to the init sequencer
`scoria_wrlvl_ifc`	the write-leveling interface: DFI leveling handshake, MR1 write path, `tWL*` window enforcement, `*_STATS` telemetry. **No search loop**

8.2 Naming, and the package

Per decision D3, scoria gets its own `scoria_pkg` with a one-bit memtype enum covering `{DDR3, LPDDR3}`. It does not extend `pumice_pkg`, whose `memtype_e` is one bit wide and already spent on `{DDR2, LPDDR2}` — widening it would change a CSR in a design that is measured and shipping, for a controller whose RTL did not exist yet. The trade was wrong in that direction then, and scoria having RTL now does not make it right: widening a shipping CSR costs the same today.

When the DDR4/LPDDR4 controller begins, a shared `mem_ctrl_pkg` with a two-bit memtype becomes worth the migration, and both existing controllers move to it together. That is the right time because the cost is paid once with three members to amortise it, rather than twice.

Note: this means `scoria_pkg` and `pumice_pkg` will contain near-identical type definitions for a period. That duplication is deliberate and time-boxed, and it is recorded here so it is not mistaken for an oversight and "fixed" by someone widening pumice's CSR.

9 The Target Design Point

9.1 Named as of v0.5

Sean, 2026-09-30: “Assume k7ddrphy on genesys2 if it helps.” It helps considerably — it converts most of this specification’s build-time parameters from open to fixed, and it gives every future bandwidth number a denominator.

The board decision for *verification* is unchanged: scoria is verified against a DFI bus functional model in simulation. What follows is the **target** the parameters are chosen for, not a claim that a bitstream exists.

Table 2.4: The target design point

Item	Value	Source
Board	Digilent Genesys 2	already on the bench for STREAM and RAPIDS
FPGA	Xilinx Kintex-7	board
PHY	`s7ddrphy.K7DDRPHY`, `memtype="DDR3"`, `nphases=4`	litex-boards `digilent_genesys2`
DRAM	2 x MT41J256M16, 32-bit bus	platform: 32 DQ, 4 DQS pairs, 4 DM
Geometry	8 banks, 32768 rows, 1024 columns	LiteDRAM `MT41J256M16`
Address pins	15 address, 3 bank	platform; matches 2^15 rows and 2^3 banks
System clock	75 MHz	owner restatement 2026-10-10 — 100 MHz was never the target (scoria BUG-003, closed)
Gear ratio	1:4	`MT41J256M16(sys_clk_freq, "1:4")`
DRAM clock	400 MHz	PHY PLL output from the 200 MHz board oscillator; not derived from sys
Data rate	800 MT/s – **DDR3-800**	2 x CK

Item	Value	Source
Theoretical peak	**3200 MB/s**	800 MT/s x 32 bit / 8

For comparison, pumice's peak is 600 MB/s (300 MT/s on a 16-bit bus). scoria's target is 5.3x that, from a wider bus and a higher rate — worth stating plainly because it changes what the front end has to sustain, not just what the DRAM can do.

Important — 3200 MB/s is the denominator. The house rule is that a bandwidth table carries the measured figure *and* the theoretical maximum. For scoria on this target the maximum is 3200 MB/s, and a percentage quoted without it is not a result. pumice learned this twice: once by publishing percentages that were understated 2x because the byte counter was per-phase, and once by quoting a figure with no peak beside it.

9.2 What this fixes

Build-time parameters that earlier editions left open now have values:

Table 2.5: Build-time parameters fixed by the target

Parameter	Value	Why it is forced
`DFI_RATE`	**4**	must equal the PHY's `nphases`, which is 4. Not a choice, and a mismatch is a functional break that presents as data corruption
`NUM_BANKS`	8	DDR3 geometry
row width	15	32768 rows. pumice's package already pads the row field to 18 bits "DDR3 forward-compat", so this fits with headroom
column width	10	1024 columns
DQ width	32	board wiring, 2 devices in parallel
`NUM_RANKS`	1	single rank per device, both in parallel on one chip select

`DFI_RATE = 4` is the happy accident worth noting: pumice also runs four DFI phases, so the inherited DFI datapath — the async-FIFO CDC, the bubble-free pipeline, the per-phase `_pN` signalling — transfers at the same gear ratio it was built and measured at. Had the target been 1:2, the datapath would still work but none of pumice's phase-level measurements would carry over.

9.3 The JEDEC timings for this part and bin

From LiteDRAM's `MT41J256M16` definition, DDR3-800 speed bin. These are the values the CSRs are initialized to; they remain **runtime** registers.

Table 2.6: DDR3-800 timings for MT41J256M16

Parameter	Value
`tREFI`	7812.5 ns (64 ms / 8192)
`tRP`, `tRCD`, `tWR`	13.1 ns each
`tRAS`	37.5 ns
`tRFC`	139 ns
`tFAW`	50 ns
`tRRD`	max(6 nCK, 10 ns)
`tWTR`	max(4 nCK, 7.5 ns)
`tCCD`	4 nCK
`tZQCS`	max(64 nCK, 80 ns)

Note: several are stated as max(nCK, ns), which is JEDEC's form and the reason the CSR derivation must take the larger of the two after converting at the operating frequency. pumice's timing derivation was corrected once for exactly this — a formula that matched one measured point but took the wrong branch elsewhere.

Warning — the cycle counts derived here are only correct at the MC clock frequency they were derived for. Every timing CSR is a count of `aclk` cycles (the controller clock at this design point is 75 MHz / 13.33 ns, restated 2026-10-10, scoria BUG-003). Change the MC clock frequency and the whole timing set — the JEDEC table above, the init and ZQ waits, the write-leveling windows — must be re-derived at the new frequency and reprogrammed before traffic resumes. A stale value violates a real DRAM timing and presents as data corruption, not as an error. There is no hardware frequency-change protocol in this generation; reprogramming is software's job.

At CK = 400 MHz (2.5 ns), the write-leveling windows of Chapter 3.3 become concrete: `tWLMRD` 40 nCK = 100 ns minimum, `tWLDQSEN` 25 nCK = 62.5 ns.

9.4 What is still open

Nothing in the specification. Two things remain *choices to be made during implementation* rather than open questions:

- **`tWLMRD`'s maximum**, which is scoria's to define as a timeout (Chapter 6). With CK at 2.5 ns a generous bound is easy to pick; the requirement is that it exists and reports distinctly, not that it be tight.
- **A bitstream, and timing closure at 75 MHz.** The board build flow now exists — board top, harness, passing lint — but no bitstream has been built. The first out-of-context synthesis (2026-10-01, scoria BUG-003) measured WNS -2.022 ns reg-to-reg against a 10 ns constraint (fmax 83.2 MHz). Against the restated 75 MHz design point (13.33 ns) that same measurement is approximately **+1.3 ns of positive slack** — the design point is met as the RTL stands. The owner restated the clock on 2026-10-10 (100 MHz was never the target; BUG-003 closed on that restatement). The OOC evidence itself is unchanged and belongs to the bug; what a real board build adds is floorplanning, congestion, and the correctly-constrained `dfi_clk` domain — a bitstream at 75 MHz remains the open item.

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10 What Is Inherited, and the Blocks That Change

10.1 The inheritance, stated once

Twenty of pumice's twenty-four FUBs carry over with no functional change, as do the top, core, AXI4 macro and the DFI datapath. Chapter 2.3 lists them; the count was twenty-one through v0.8, and v0.9 moves `refresh_ctrl` to MODIFIED — the `REFpb` inheritance stands, but TASK-001's Modes A/B are a real change to the block, specified in Ch 3.4. This chapter covers only what changes, and the list is short by design.

The inheritance is an argument, not a convenience. Those blocks have been through board bring-up, a read-path rebuild that took reads from 291.7 to 571.3 MB/s, and a formal campaign across nine modules. Re-deriving them would discard that evidence and reintroduce the bugs it caught.

Important — inherit `global_timers` in its FIXED form. pumice's version published readiness one cycle late: the registered status flop sampled the current counter state rather than the next, so the flags alone permitted tCCD and tRTW violations. The fix computes one next-state function and feeds both the counter and its status flop from it. A fresh implementation written from a behavioural description of the same block would reintroduce the defect, because the description does not mention the flop. Copy the module.

10.2 The changes, in one table

Table 3.1: Every change, with its cause

Block	Marking	Cause	Where
`init_sequencer`	MODIFIED	RESET# is a pin; MR order is MR2-MR3-MR1-MR0; ZQCL closes init	3.2
`mode_register`	MODIFIED	MR0-MR3 replaces MR0-MR2 plus EMRS3	3.2
`dfi_cmd_formatter`	MODIFIED	new encodings: ZQCL, ZQCS, PREA	3.2
`scoria_zq_ctrl`	**NEW**	periodic ZQCS is maintenance traffic; carries the CSR-selectable placement policy (TASK-001 Mode C)	3.2
`scoria_wrlvl_ifc`	**NEW**	DDR3 adds write leveling	3.3
`refresh_ctrl`	MODIFIED (v0.9)	the REFpb mechanism is inherited unchanged, but TASK-001 Modes A/B add elastic pull-in/postpone streaks and the TCR tREFI derate behind CSRs; JEDEC ±8 ceiling untouched	3.4
`powerdown_ctrl`	INHERITED	mechanism unchanged: CKE +	below

Block	Marking	Cause	Where
		SRE/SRX. The DFI low-power channel is unused on this PHY family	
`dfi_signal_pack`	INHERITED	every signal it packs is identical in v2.1.1 and v3.1 for DDR3; v3.1's new channels are driven at the plane above	Ch 4.1
`scoria_scheduler_plane`	MODIFIED	must admit ZQ demand alongside refresh	3.2
`scoria_csr`	MODIFIED	new timing registers and leveling telemetry; TASK-001 mode-select fields and `REF_STATS` telemetry	Ch 5

10.3 Power-down: smaller than v0.1 thought

DDR3 and LPDDR3 both offer precharge-power-down, active-power-down and self-refresh (SRE / SRX). pumice's `powerdown_ctrl` already implements idle-timer-driven entry and exit, and **that is the whole mechanism** on a Series-7 PHY target.

v0.1 through v0.3 of this document said the interesting change was that DFI v3.1 splits `dfi_lp_req` into `dfi_lp_ctrl_req` and `dfi_lp_data_req`, and left the exit-ordering as an open question. The split is real, but it is not load-bearing here: **`s7ddrphy` implements no DFI low-power interface at all**, and LiteDRAM's generated DDR3 controller does not drive one either — zero `dfi_lp` hits in the generated core against 92 references to `CKE`.

Power-down on this PHY family is a **DRAM command** matter: CKE, plus SRE and SRX for self-refresh. `powerdown_ctrl` is therefore INHERITED in mechanism, and what it gains for DDR3 is the new self-refresh command encodings rather than a new interface.

Status, 2026-09-30: not in the design, by decision (Sean). The module is INHERITED as a FILE; nothing instantiates it, and `scoria_top` has no CKE port at all – so the feature is absent rather than unconnected, and the DRAM simply never leaves normal operation. That matches the reference: LiteDRAM's generated DDR3 core for this board has **no power-down or self-refresh engine of any kind**. Its only `power_down` references are the MMCM's `PWRDWN` pin, and the 92 `CKE` references resolve to a CSR bit – `main_litedramcore_sdram_cke = main_litedramcore_sdram_storage[1]`, fanned identically to all four phases – i.e. software raises CKE during init and it stays raised. There is no SRE/SRX issue path.

So scoria holding CKE high for ever is parity with the controller that passes memtest on this hardware, not a shortfall against it. The same applies to `dfi_signal_pack`, also INHERITED-as-a-file and also uninstantiated: every DFI bus here is phase-multiplied by construction (`DFI_ADDR_BUS_W = ROW_WIDTH * DFI_RATE`, and so on) and LiteDRAM's `dfi_p0..p3_*` are per-phase for the same reason, so there is no packing step for either design to perform.

That justification is DDR3-scoped, and does not transfer to LPDDR3. The argument above is parity with a DDR3 reference; LiteDRAM's DDR3 core says nothing about what an LPDDR3 part needs. LPDDR exists FOR power: self-refresh is a mobile part's normal idle state rather than an exotic mode, and LPDDR3 adds **Deep Power Down**, which DDR3 does not have – `scoria_pkg` already marks `OP_DPDE` "LPDDR3 only". DPD requires stopping the DRAM clock, which is `dfi_dram_clk_disable_o`, and that signal is `dfi_signal_pack`'s one function NOT covered by the phase-multiplied buses. `powerdown_ctrl`'s own v3 TODO names the dependency: DPD entry "needs `dfi_dram_clk_disable_o` cooperation from scoria_dfi_signal_pack". So for LPDDR3 the two are a PAIR, and both are load-bearing.

What stands in the way is four gaps, not one wire:

Level	State today
`powerdown_ctrl` / `dfi_signal_pack`	not instantiated

Level	State today
`scoria_top` / `scoria_dfi_plane`	no `dfi_cke` or `dfi_dram_clk_disable` port at all
`scoria_dfi_cmd_formatter`	`OP_SREFE` / `OP_SREFX` / `OP_DPDE` fall to the `default:` arm and are driven as NOP
`scoria_cmd_arbiter`	never picks any of those three ops

So the decision stands for the DDR3 target and must be re-made on LPDDR3's own terms the moment an LPDDR3 device is in scope. It also tips the housekeeping question: these two files are the starting point for that work, so keeping them dormant is worth more than deleting them. See scoria-ddr3-lpddr3 TASK-003.

Requirement for the DFI low-power ports. scoria still exposes `dfi_lp_ctrl_req` and `dfi_lp_data_req`, because DFI v3.1 defines them and a future PHY may consume them. Their behaviour when no acknowledgement ever arrives is specified: **time out and report; never block power-down.** An unacknowledged request must not be able to wedge the controller — which is the real hazard, and a smaller one than the ordering puzzle v0.2 imagined.

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11 Init Sequencer: RESET#, MR0-MR3, ZQ Calibration

11.1 The sequence is JEDEC's, verbatim

JESD79-3F specifies DDR3 power-up initialization as a numbered sequence. It is reproduced here as *requirements on the sequencer*, in the spec's order, because the order is not ours to choose:

Table 3.2: DDR3 initialization requirements, from JESD79-3F

Step	Requirement
1	Power ramp, with the spec's VDD / VDDQ / VTT / Vref ordering constraints. Board-level, not controller-level
2	After `RESET#` is de-asserted, wait **500 us** before CKE becomes active. The DRAM initializes internally during this, independent of external clocks
3	Clocks stable for at least **10 ns or 5 tCK, whichever is larger**, before CKE goes active. A NOP or Deselect must be registered before CKE goes high, meeting tIS
4	CKE must be **continuously registered high** from that point until initialization finishes — including expiry of `tDLLK` and `tZQinit`
5	Wait `tXPR` = max(`tXS`, 5 x tCK) after CKE high before the first MRS
6	MRS to **MR2**
7	MRS to **MR3**
8	MRS to **MR1**, with DLL enabled
9	MRS to **MR0**, with DLL reset
10	**ZQCL** to start ZQ calibration
11	Wait for both `tDLLK` and `tZQinit`
12	Ready for normal operation

11.2 What DDR3 does NOT need, and DDR2 did

Found while implementing the sequencer, and worth stating because it makes the block smaller rather than larger: the DDR3 sequence has **no precharge-all, no double auto-refresh, no OCD default/exit pair, and no second MR0 load**.

pumice's DDR2 sequencer has all four. The second MR0 load existed only to clear the DLL-reset bit; DDR3 does not need it because that bit is **self-clearing** (JESD79-3F 3.4.2.4). So scoria's init FSM drops seven states relative to pumice's and adds three (RESET# hold, the 500 us CKE wait, tXPR), for a net reduction.

11.3 Three consequences for the sequencer

RESET# is a pin, not a command. This is the structural change. pumice's init sequencer issues commands; scoria's must additionally drive a reset output and hold it through a timed window. That output has to reach the top level and the PHY, and it is the first thing in the sequence rather than something the command path can express.

The MR order is MR2, MR3, MR1, MR0 — and it is not sorted. Steps 6 through 9 are explicit, and MR1 carries "DLL enable" while MR0 carries "DLL reset", so the order encodes a dependency rather than a preference.

Note: pumice hit precisely this. Its init programmed EMRS3 before EMRS2; the error was benign in practice but wrong, and it was corrected to JEDEC's order. DDR3's MR2-MR3-MR1-MR0 is the same pattern extended to a four-register set, and the same rule applies: the sequence is a citation, not a design choice.

ODT is constrained during init. The DRAM holds on-die termination in high-impedance while RESET# is asserted and until CKE is registered high; thereafter ODT must be held statically — and **statically LOW if `RTT_NOM` is to be enabled in MR1** — until initialization completes. The sequencer therefore owns ODT during init, not just during traffic.

11.4 ZQ calibration

ZQCL closes initialization (step 10) and tZQinit gates readiness (step 11). Outside init, DDR3 offers ZQCS, the short calibration, and tZQoper for a long one. All three gate command issue alongside tMRD, tMOD and tRFC.

11.4.1 Why ZQ gets its own block

ZQCL at init belongs to the sequencer. Periodic ZQCS does not, because it is **maintenance traffic**: the controller must issue it on an interval no host asked for, competing with demand traffic for the command bus. That is the same shape as refresh, and refresh has its own controller for the same reason.

scoria_zq_ctrl therefore:

- holds the ZQCS interval as a **runtime CSR**, so it can be swept;
- raises a demand to the scheduler rather than seizing the bus;
- enforces tZQCS before releasing the bus back;
- reports a count of issued calibrations, so the interval can be confirmed from the host rather than assumed;
- carries a CSR-selectable **placement policy** (ZQ_CFG.placement, TASK-001 Mode C, specified below).

The scheduler change is one input, not a rework. scoria_scheduler_plane already arbitrates refresh demand against demand traffic; ZQ adds a second maintenance source with the same shape.

Placement policy (v0.9, TASK-001 Mode C). Whether ZQCS should *preempt* queued demand traffic, or only fill gaps, is no longer an open question — it is a CSR-selectable policy, reset to the baseline. ZQ_CFG.placement selects: 0 = request on interval expiry (the baseline above); 1 = defer-under-demand — at expiry, if the scheduler reports sustained demand, the request is held rather than raised, up to ZQ_CFG.overdue_max MC cycles (0 = uncapped), after which the request fires regardless. Policy value 2 is reserved. Two things hold in every policy: the module still **requests and waits — it never preempts** (the Ch 6 Q2 answer stands), and tZQCS is still enforced by holding off the next interval rather than blocking traffic. The starvation revisit condition named in Ch 6 Q2 — the interval slipping materially past its CSR value under sustained demand — is now measurable with the instruments this block already carries (obs_overdue, obs_zqcs_total) plus the overdue_max cap itself, which turns runaway deferral from a suspected hang into a bounded, observable policy choice. Per-mode design detail is in design-requirements.md §6.

12 Write Leveling: the Interface, Not the Search

12.1 What write leveling is for

On a fly-by command/address topology, DQS arrives at each DRAM at a different time relative to CK. Write leveling lets the controller discover that skew per device: the DRAM is put into a leveling mode via MR1, the controller drives a DQS edge, and the DRAM reports on its prime DQ bit whether that edge arrived before or after the CK rising edge. The controller walks its DQS delay until the reported value flips. That flip is the answer.

12.2 The decision: scoria provides the interface, firmware runs the search

This was settled as decision D2. `scoria_wrlvl_ifc` contains **no search loop**.

The reasoning is the family's standing architectural boundary. A delay-line walk is PHY calibration: the step size, the number of taps, their monotonicity and their temperature behaviour are all properties of the PHY, not of DDR3. pumice made the same call and levels its board from firmware over a CSR passthrough, with no hardware state machine — thirteen PHY knobs reached indirectly, the search in a host program, the settled result cached.

The alternative — a hardware FSM — buys independence from a host at the cost of importing PHY-specific search logic into a portable controller, and of a calibration bug being a silicon respin instead of a script edit.

Important: this is a boundary decision, not an efficiency one. A future system that needs autonomous leveling should add an external calibration engine driving this interface, not a search loop inside the controller.

Corroborated by LiteDRAM. A DDR3 LiteDRAM core was generated to check this (`bin/litedram_ddr3_ref.yml`). Its write-leveling support is two CSR fields – `wlevel_en` to enter the mode and `wlevel_strobe` to emit a DQS pulse – and **no state machine whatsoever**: a search for leveling FSM state in the generated core returns nothing. Software enables the mode, pulses the strobe, reads the result and walks the delay.

An independent and widely-deployed controller reaching the same conclusion is worth more than the argument above on its own. It also tells us the *shape* of the interface that works in practice: enable plus strobe, rather than a higher-level "do a leveling pass" command.

12.3 What the block owes the system

Table 3.3: `scoria_wrlvl_ifc` responsibilities

Responsibility	Detail
DFI leveling handshake	drive `dfi_phylvl_req_cs_n` and observe `dfi_phylvl_ack_cs_n`, with `dfi_phy_wrlvl_cs_n` selecting write leveling; per chip select, per DFI v3.1
MR1 path	enter and leave the DRAM's write-leveling mode by MRS to MR1, sequenced against the handshake
Timing windows	enforce `tWLMRD`, `tWLDQSEN`, `tWLO` and `tWLOE` as **runtime CSRs**
Result capture	present the prime DQ bit's reported value to the host, per chip select
Telemetry	`*_STATS`: attempts, flips observed, current window, and whether a leveling pass has ever completed

12.4 The timing windows

From JESD79-3F:

Table 3.4: Write-leveling timing windows

Parameter	Value	What the controller must do
`tWLDQSEN`	25 nCK minimum	wait this long after entering leveling mode before driving DQS low / DQS# high; by its expiry the DRAM has applied ODT to those signals
`tWLMRD`	40 nCK minimum	wait this long before the first DQS pulse. **The maximum is controller-dependent** — the spec explicitly declines to bound it
`tWLO`	per speed bin	the DRAM returns the result on the prime DQ bit(s) asynchronously, this long after the DQS edge
`tWLOE`	per speed bin	DQ output uncertainty, defined to permit mismatch across DQ bits

What the PHY does during leveling, and what it does not. `s7ddrphy` exposes exactly `_wlevel_en` and `_wlevel_strobe`, and while leveling is enabled it forces `rddata_valid` on every phase (so the DRAM's answer returns through the ordinary read path) and forces `dqs_oe` (so DQS is driven continuously rather than gated per burst). It contains **no timeout of any kind** — which is why the maximum below is scoria's to define and not a number to be read off a PHY.

Note on `tWLMRD`'s open maximum. Because JESD79-3F leaves the maximum to the controller, scoria must *define* one rather than inherit it, and the definition belongs in the CSR: a host that waits forever for a result it will never get is indistinguishable from a broken link. The requirement is a CSR-settable timeout with a distinct status bit, so a leveling attempt that never reports is reported as a timeout and not as a flip that failed to arrive.

Note on `tWLOE`. It exists to allow mismatch *across DQ bits*, which only matters if the controller samples more than the prime bit. If the implementation samples one bit, `tWLOE` is inert — and that should be an explicit, documented choice rather than an accident.

12.5 The telemetry requirement, and why it is not optional

The *_STATS surface is a requirement rather than a nicety, and the reason is a lesson paid for on pumice's board: a detector that has never fired is not evidence of anything. pumice's data-integrity CRC checker had never once fired, and proving it worked required deliberately corrupting the analog eye until it reported every beat mismatched.

A leveling interface that reports "converged" without exposing the attempts and the observed flips is the same trap. The requirement is therefore that a host can distinguish four states: never attempted, attempted and converged, attempted and timed out, and attempted with no flip found across the whole window.

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13 Refresh: DDR3 All-Bank and LPDDR3 Per-Bank

13.1 Marking: MODIFIED (v0.9) — inherited mechanism, TASK-001 modes on top

This chapter's marking has now changed twice. v0.1 marked `refresh_ctrl` MODIFIED on the assumption that LPDDR3 per-bank refresh was new work. v0.2 corrected it to INHERITED, because **pumice already implements it**, for LPDDR2, and the mechanism LPDDR3 specifies is the same one. v0.9 corrects it once more, to MODIFIED: the inherited mechanism described below is unchanged, but TASK-001's Modes A and B — demand-aware elastic refresh and temperature-compensated refresh — are a named, bounded change planeed on top of it. They are scheduling policy only; the JEDEC ±8 credit ceiling is untouched. The modes are specified at the end of this chapter.

13.2 What pumice already has

`refresh_ctrl` carries the retention interval, all-bank refresh, JEDEC's plus-or-minus-eight postpone and pull-in scheduling — proven in formal across all sixteen postpone values — **and a REFpb bank rotor**.

The rotor is the part worth understanding, because it is not a scheduler:

```
r_bank_rotor advances exactly when a REFpb COMMAND is granted onto the
wire,
and HOLDS through REFab mode.
```

It mirrors the **device's** internal bank counter. pumice's own comment cites JESD209-2 §6.6 for why: the REFpb command *carries no bank address*. The controller cannot nominate a bank. It can only track which bank the device will refresh next, so that the bank timers know which bank is about to become busy.

The rotor also deliberately holds across a mode change rather than resetting, because the device's counter persists — and clearing the mirror would desynchronise it from the device.

13.3 LPDDR3 keeps the same mechanism

JESD209-3C specifies a bank counter in the memory device, and states that the bank sequence for per-bank refresh is **fixed to be a sequential round-robin**. During a REFpb operation, banks other than the one being refreshed remain accessible.

That is the LPDDR2 mechanism unchanged, so the pumice implementation transfers.

Important — this retires an open question. v0.1 of this document asked (as Q5) whether the REFpb round-robin order should be strictly sequential or follow bank occupancy. **The question was malformed.** Occupancy-aware ordering is not available at any price: the command carries no bank address and the device's sequence is fixed. There is nothing to choose. The question is struck rather than deferred, because deferring it would imply a decision still exists.

13.4 Refresh by memtype

Table 3.5: Refresh by memtype

Memtype	Refresh available
DDR3	all-bank `REF` only, with the JEDEC postpone window. DDR3 adds no controller-directed per-bank refresh
LPDDR3	all-bank `REF`, plus `REFpb` in the device's fixed sequential order

Per-bank refresh matters because an all-bank refresh blocks every bank at once while `REFpb` leaves the others accessible. On a workload with bank parallelism that is the difference between a refresh being a stall and being nearly free.

The cleverer schemes — out-of-order per-bank refresh, write-refresh parallelisation, refresh pausing, SARP and DSARP — are LPDDR4/DDR5 or research, and scoria TASK-001 assigns them to the DDR4 area rather than here. That split stands.

13.5 Requirements

1. Offer all-bank refresh with the JEDEC postpone/pull-in window (inherited), and for LPDDR3 additionally `REFpb` with the rotor mirroring the device's counter (inherited).
2. The all-bank versus per-bank choice on LPDDR3 is a **runtime CSR**, so both can be measured from one bitstream. This is the only genuinely new surface in this block, and it is a mode select rather than a mechanism.
3. **Re-establish the retention property for the per-bank path.** The inherited formal proof covers retention headroom across every postpone value for all-bank refresh. `REFpb` changes the arithmetic that proof checks — the interval seen by any one bank is the all-bank interval multiplied by the bank count — so the property must be re-derived for the per-bank mode.
4. `REFpb` must respect per-bank timing as an ordinary occupant of that bank: a per-bank refresh is an operation on a bank and the bank timer owns it.

Important: requirement 3 is the one most likely to be skipped, and it is the reason refresh can be trusted at all. Carrying the all-bank property forward unchanged would prove the wrong thing while appearing green — the failure mode this family has been bitten by before.

Note: pumice's idle detection is inherited with the block and is worth knowing about, because it looks like an arbitrary constant. Demand is CAM occupancy, which blinks off for a few cycles between bursts; treating those micro-gaps as idle would release postponed refreshes mid-stream and trigger pull-ins. Only a sustained gap counts — confirmed over sixteen cycles in the inherited baseline, and over the CSR-configurable `pullin_idle_streak` (reset 16, the same value) once Mode A below is enabled.

13.6 What TASK-001 adds: Modes A and B

Both modes are CSR-selected, reset to off, and bit-identical to the baseline above when disabled — the same "config not param" rule the rest of the controller runs on. Per-mode design detail lives in `design-requirements.md` §6; only the architecture-relevant contract is restated here.

13.6.1 Mode A — demand-aware elastic refresh

`REF_CTRL.elastic_en` (reset 0 = baseline), with two threshold fields in the same register: `pullin_idle_streak` (reset 16 — the inherited constant, now sweepable) and `postpone_demand_streak` (reset 1).

- **Pull-in** fires only after demand has been idle for `pullin_idle_streak` MC cycles, generalizing the inherited sixteen-cycle confirmation into a host-sweepable threshold.
- **Postpone** engages only under a sustained demand streak of `postpone_demand_streak` cycles. Below that, demand counts as sporadic and the strict baseline stands: a refresh is requested on every interval expiry while anything is pending.
- The JEDEC ±8 postpone/pull-in credits remain the hard ceiling in every configuration; Mode A never touches the credit arithmetic.
- Telemetry: `REF_STATS_POSTPONE` / `REF_STATS_PULLIN` count the events, so an elastic-refresh sweep is measurable in-system rather than inferred.

13.6.2 Mode B — temperature-compensated refresh (TCR)

`REF_CTRL.tcr_en` (reset 0) + `trefi_derate` (reset 0 = 1x; 1 = 2x, 2 = 4x; the illegal encoding 3 clamps to 4x). Firmware writes the JESD79-3F §7.3.3 refresh-rate class and the tREFI tick scales with it — a cheaper refresh rate where temperature allows, at a retention margin the temperature guarantees. Three properties matter:

- Derate applies to the **counter reload only**: a mid-interval CSR change takes effect on the next reload; the running counter is not rescaled.
- `TEMP_DERATE_RANK0` keeps its LPDDR3-MR4 hardware-written path untouched for the future LPDDR3 build; the DDR3 path is this software-written select.
- The retention property is **re-derived with the derate factor in the interval arithmetic** — the same rule as `REFpb` above: a changed interval changes the proof, and the proof is never carried forward green.

Formal status: with both modes present, the retention and accounting properties remain green in `formal/scoria/refresh_ctrl`, with cover points at the configured pull-in streak boundary (evidence: scoria TASK-001 close, 2026-10-03).

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14 DFI v3.1 Master Interface, and the v2.1.1 Delta

14.1 The revision decision

scoria targets **DFI v3.1** (decision D1). pumice's DFI plane was built to v2.1.1, so this chapter is the delta between them, measured across the two specifications: 126 distinct `dfi_*` signal names in v2.1.1, 151 in v3.1.

The decision was not obvious and the reasoning is on the record: v3.1's *gain* for a DDR3/LPDDR3 controller is the leveling rework, the low-power split and `dfi_init`. Everything else it adds is DDR4/LPDDR4. Building to v2.1.1 and hand-adding write leveling would reuse pumice's plane more directly. v3.1 wins because the per-CS leveling scheme is what the DDR4 controller will need too, and paying for it once here is cheaper than paying twice.

14.2 What does NOT change, and a diff that lies about it

The datapath is inherited entirely. One async-FIFO CDC, bubble-free, the unit of transfer a DFI word, `DFI_RATE = 4`.

Important. A naive signal-name diff between the two specifications reports `dfi_address_p2`, `dfi_cs_n_p3`, `dfi_bank_n_p2`, `dfi_odt_p3` and their siblings as **removed in v3.1**. They are not removed. v2.1.1 enumerates phase variants explicitly; v3.1 generalises the notation to `_pN` — `dfi_address_pN`, `dfi_cs_n_pN`, `dfi_odt_pN`, `dfi_bank_pN` — and shows `_p0`/`_p1` only as examples. The 1:4 frequency ratio remains a defined mode (v3.1 §3.2.4, §3.3.4).

pumice runs four phases. An implementer who trusted that diff would rewrite a working, measured datapath for no reason, so the trap is recorded here rather than left for them to fall into.

14.3 What changes and matters

14.3.1 Leveling: reworked outright

Table 4.1: The leveling interface, v2.1.1 against v3.1

v2.1.1	v3.1
`dfi_rdlvl_mode`, `dfi_rdlvl_load`, `dfi_rdlvl_delay*`, `dfi_rdlvl_gate_mode`	**removed**
`dfi_wrlvl_mode`, `dfi_wrlvl_load`, `dfi_wrlvl_delay*`	**removed**
`dfi_rdlvl_cs_n`, `dfi_wrlvl_cs_n`	**removed**
—	`dfi_phylvl_req_cs_n`, `dfi_phylvl_ack_cs_n` (request/ack, per chip select)
—	`dfi_phy_wrlvl_cs_n`, `dfi_phy_rdlvl_cs_n`, `dfi_phy_rdlvl_gate_cs_n` (which leveling)
—	`dfi_lvl_pattern`, `dfi_lvl_periodic`

This is the change scoria cannot route around, because DDR3 adds write leveling. `scoria_wrlvl_ifc` (Chapter 3.3) drives this interface, and because the search is firmware-driven the block is an interface and a set of timing windows rather than a state machine.

14.3.2 Low power: one request becomes two, and nothing consumes either

`dfi_lp_req` splits into `dfi_lp_ctrl_req` and `dfi_lp_data_req`, letting the command path and the data path be put to sleep independently.

In practice, on this PHY family, the channel is unused. `s7ddrphy` implements no DFI low-power interface, and LiteDRAM's generated DDR3 controller drives none. Nor does it power down by DRAM command: measured in the generated core, there is no self-refresh or power-down engine at all, and CKE is a CSR bit software raises during init. scoria exposes both requests for spec compliance and for a future PHY that consumes them, with the specified behaviour that an unacknowledged request times out and reports rather than blocking power-down. See Chapter 3.1.

14.3.3 Initialization and errors

`dfi_init` appears, giving initialization its own interface — which pairs with DDR3's RESET# pin. `dfi_parity_error` becomes `dfi_error` plus `dfi_error_info`, a wider surface that mostly

carries DDR4 CA-parity detail; scoria uses the error indication and ignores the DDR4-specific encodings.

14.3.4 Reset and the data-phase chip selects: presented, and how

Two surface details are worth recording exactly, because the DV framework's DFI signal catalog checks the required set for the declared version and names what is absent (scoria TASK-005 was filed by exactly that check):

- **`dfi_reset_n`.** DFI has carried DRAM RESET# on the control interface for DDR3 since v2.1 — the signal is gated on memory type, not version — so any scoria DFI claim, not just the v3.1 one, implied this name. scoria drives the signal from the init sequencer, and on the DFI-presenting modules it is port-named `dfi_reset_n_o`. Inside the controller it remains `dram_reset_n_o`: there it is device reset sequencing, and it becomes a DFI signal at the module that presents the DFI bus.
- **`dfi_wrdata_cs_n` / `dfi_rddata_cs_n`.** Genuinely v3.1-only (min 3.1, max 3.1): the per-data-phase chip selects, added so a multi-rank system can say which rank owns the DQ bus during the data window — and, during training, which rank is under training. scoria drives both. At `NUM_RANKS = 1` the only legal value is rank 0, and both are driven constant `'0`; the CS-under-training role of `dfi_wrdata_cs_n` is degenerate at one rank, where the rank being levelled is always CS0. A multi-rank build must drive them from the granted command's rank, in step with `dfi_cs_n`.

14.4 What scoria does NOT implement

Named explicitly so their absence is a decision on the record:

Table 4.2: DFI v3.1 features deliberately unimplemented

Group	Signals	Why not
DDR4 command encoding	`dfi_act_n*`, `dfi_bg*`, `dfi_cid*`	DDR3 has no dedicated ACT pin, no bank groups, no chip ID
CA parity, CRC, alert	`dfi_alert_n*`, `dfi_parity_in_p*`	DDR4-era
Data Bus Inversion	`dfi_rddata_dbi*`	DDR4/LPDDR4
CA training	`dfi_calvl_*`, `dfi_ca_capture`, `dfi_phy_calvl_cs_n`	LPDDR4-era

That is the majority of v3.1's growth over v2.1.1. Targeting the newer revision does not mean implementing all of it.

15 AXI4 Slave and APB CSR

Both interfaces are **INHERITED** from pumice without functional change. This chapter records what that inheritance includes, so that “inherited” is not mistaken for “unspecified”.

15.1 AXI4 slave

The host-side interface is pumice's `axi4_ifc`: one read intake and one write intake at 1:1, a write-data CAM, a read-reorder CAM, in-order commit, and FSM-free split and aggregate where the aggregate and last flags ride with the command in the CAM.

Nothing about DDR3 or LPDDR3 reaches the host side. The DRAM generation is invisible across this boundary, which is the property that makes the family worth having.

Width gearing is inherited too. The host bus width need not match the DRAM burst width: formally-verified dwidth converters wrap the core, so the host side is free. This is why `scoria_top_geared` exists as a separate top alongside `scoria_top`.

15.2 APB CSR

A PeakRDL-generated register block behind an APB slave. The register *set* grows (Chapter 5), but the interface and the generation flow do not change.

Important — the regeneration rule applies unchanged. The register block is generated through `bin/peakrdl_generate.py`, never raw `peakrdl`, and always with an explicit `-o`. The wrapper emits the RTL, the documentation and the Python regmap together; raw peakrdl skips the latter two and the default output path can write an orphan copy the build never reads, which then drifts.

Important — registers are accessed by NAME. Host code reads and writes through the generated regmap, never by hardcoded offset. Offsets churn whenever the RDL changes, and a host that hardcodes them fails silently by hitting a neighbouring register that happens to decode.

15.3 The CDC on the CSR path, and why it is an async FIFO

If scoria is integrated the way pumice is on its board — an APB window on one reset domain reaching a core on another — then the CDC between them must be a gray-pointer async FIFO, not a toggle-parity handshake.

The reason is an asymmetric reset, and it is worth carrying forward because it cost real debugging on pumice: the APB side runs on `presetn` and the core side on `aresetn`, and a soft reset pulses **only the core side**. A toggle-parity handshake desynchronises across that and leaves the response stream permanently offset by one transaction — every readback returns the answer to the previous question, which looks like a lagging register file rather than a CDC fault.

This is a note about integration rather than about scoria's own ports, but it belongs here because the next person to simplify that CDC will be reading this chapter.

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16 scoria_pkg, and Build-Time vs Runtime

16.1 The package (decision D3)

scoria gets its own `scoria_pkg`, with a one-bit memtype enum:

```
typedef enum logic {
    MEMTYPE_DDR3   = 1'b0,
    MEMTYPE_LPDDR3 = 1'b1
} memtype_e;
```

It does **not** extend `pumice_pkg`. That package's `memtype_e` is one bit wide and already spent on `{DDR2, LPDDR2}`; adding two more members means widening the field, which reaches pumice's RTL and its CSR — a measured, shipping design — for the benefit of a controller whose RTL did not exist yet. The trade was wrong in that direction then, and scoria having RTL now does not make it right: widening a shipping CSR costs the same today.

When this changes. At the start of the DDR4/LPDDR4 controller, a shared `mem_ctrl_pkg` with a two-bit memtype becomes worth the migration, and both existing controllers move to it in the same pass. That is the right moment because the cost is paid once with three members to amortise it.

Note: until then `scoria_pkg` and `pumice_pkg` carry near-identical type definitions. The duplication is deliberate and time-boxed. It is recorded here so that a later reader does not "fix" it by widening pumice's CSR — which is precisely the thing this decision avoids.

Inherited from pumice's package: the row field is already 18 bits wide, with the comment "DDR3 forward-compat". That was written in anticipation of this controller and carries over as-is.

16.2 The rule: timings are runtime, geometry is build-time

Inherited from pumice and restated because it is the single most load-bearing convention in the family:

Every enforced timing is a runtime CSR. Not a parameter. A timing compiled in cannot be swept, and a controller whose timings cannot be swept cannot be characterized — which is how a controller ends up with numbers nobody can explain.

Corollary — a runtime timing CSR is a count of `aclk` cycles, so its value is bound to the MC clock frequency. "Runtime" does not mean "frequency-independent": the register holds cycles, not nanoseconds, and the correct count changes when the clock does. A frequency change — retargeting the build to a different `aclk` for timing margin, for example — invalidates every cycle-counted CSR at once, and software must re-derive the whole set from the JEDEC ns/nCK values before traffic resumes. Stale values violate real DRAM timings and present as data corruption, not errors.

Build-time is reserved for what cannot be runtime: bus widths, the DFI gear ratio, CAM depths, bank and rank counts. The test is whether changing it changes the amount of hardware.

As of v0.5 these have values, fixed by the target design point in Chapter 2.4: `DFI_RATE = 4` (the PHY's `nphases`), `NUM_BANKS = 8`, row width 15, column width 10, DQ width 32, `NUM_RANKS = 1`. The DDR3-800 JEDEC timing set for the part is tabulated there too, and those remain runtime CSRs initialized to those values.

Table 5.1: Build-time versus runtime

Kind	Build-time or runtime	Examples
Bus and geometry	build	data width, address width, `NUM_BANKS`, `NUM_RANKS`, CAM depths
DFI gearing	build	`DFI_RATE`, which must equal the PHY's phase count
Memtype	build	`MEMTYPE_DDR3` / `MEMTYPE_LPDDR3`
JEDEC timings	**runtime**	every command-spacing and recovery parameter
New DDR3 timings	**runtime**	`tZQinit`, `tZQoper`, `tZQCS`, `tWLMRD`, `tWLDQSEN`, `tWLO`, `tWLOE`
Policy selects	**runtime**	page policy, refresh

Kind	Build-time or runtime	Examples
		mode (all-bank vs `REFpb`), ZQCS interval, and the TASK-001 mode selects — elastic refresh, TCR derate, ZQCS placement — with their thresholds (idle/demand streaks, `overdue_max`)

Important — `DFI_RATE` must equal the PHY's phase count, set identically on both sides. This is inherited and it is not negotiable: a mismatch is not a performance problem, it is a functional break, and it is the kind that presents as data corruption rather than as an error.

16.3 New CSR groups

The register set grows by three groups. Exact offsets are not fixed in this edition — they come from the RDL, which is authored with the RTL — but the content is specified:

Table 5.2: New CSR groups

Group	Contents
ZQ	`tZQinit`, `tZQoper`, `tZQCS`, the ZQCS interval, and a count of calibrations issued
Write leveling	`tWLMRD` (with its controller-defined maximum as a timeout), `tWLDQSEN`, `tWLO`, `tWLOE`, per-CS result, and the `*_STATS` telemetry of Chapter 3.3
Refresh mode	all-bank versus `REFpb` select, for LPDDR3
Advanced modes (TASK-001)	the elastic-refresh and TCR fields in `REF_CTRL`, placement/`overdue_max` in `ZQ_CFG`, and the `REF_STATS_POSTPONE` / `REF_STATS_PULLIN` telemetry — all specified in Chapters 3.2 and 3.4

Note: `tWLMRD`'s maximum is a *scoria* number, not a JEDEC one — JESD79-3F declares it controller-dependent. It appears in the CSR as a timeout with its own status bit, so that a leveling attempt which never reports is distinguishable from one that reports no flip.

17 Verification Strategy, and the Open Questions

17.1 The target is the DFI boundary

Verification is against the DV repository's DFI bus functional model, in cocotb. No board is required to verify scoria, and that has not changed.

A target board is now named, though (Sean, 2026-09-30, Chapter 2.4): the Genesys 2 with `K7DDRPHY` and two MT41J256M16 devices at DDR3-800. That fixes the parameters the DUT is verified *with*, so the BFM should be configured to that geometry – 8 banks, 32768 rows, 1024 columns, a 32-bit bus, four DFI phases – rather than to an arbitrary one. Verifying at the geometry the design point uses is free, and verifying at a different one silently weakens every result.

That boundary is chosen for the same reason pumice chose it: it is the one interface where the controller's obligations are fully specified by a standard, so a model can be authoritative rather than approximate. pumice's experience is that code passing against the BFM passes on hardware modulo PHY training — which is exactly the residue this specification pushes into firmware anyway.

17.2 What must be verified, beyond inheriting pumice's suites

The inherited blocks bring their tests with them. The new and modified ones need new coverage, and four items are specific enough to name now:

1. **The init sequence, in JEDEC's order.** Not "the DRAM came up" but step-by-step: `RESET#` held, the 500 us wait, clocks stable before CKE, CKE continuously high, `tXPR` before the first MRS, then MR2, MR3, MR1, MR0, then `ZQCL`, then `tDLLK` and `tZQinit`. A checker that asserts the *order*, because order is what pumice got wrong on its equivalent.
2. **ODT during init.** High-impedance while `RESET#` is asserted, and held statically — LOW if `RTT_NOM` is enabled in MR1 — until init completes.
3. **Write leveling as a protocol, not an outcome.** The handshake sequence and every timing window, with the `tWLMRD` timeout exercised deliberately. A leveling interface whose timeout path has never run is an untested path on the only path that reports failure.
4. **Per-bank refresh retention.** The inherited formal property proves retention headroom across all sixteen postpone values for all-bank refresh. `REFpb` changes that arithmetic — the per-bank interval is the all-bank interval times the bank count — so the property must be re-established, not carried forward. Carried forward unchanged it would prove the wrong thing and appear green.

Important — the formal area is the right home for the spacing properties. No assertions go in the RTL. pumice's `formal/mc_common` area proved nine modules and found two real bugs, including the `global_timers` next-state defect this specification tells scoria to inherit the fix for. scoria should open its own formal area early, and the `cmd_history_checker` — which independently re-derives JEDEC spacing from the issued command stream — should grow DDR3's parameters before the RTL is trusted.

17.3 Status, 2026-10-03

The controller this chapter describes now exists. Measured 2026-10-01 (scoria BUG-003, which quotes the full regression as evidence): **221 tests passing, 9 formal blocks**, and the top tier runs AXI in and DFI out with data through the whole datapath. The init sequencer, ZQ controller and write-leveling interface of Chapter 3 are implemented and covered; the two dormant inherited FUBs (`powerdown_ctrl`, `dfi_signal_pack`) are retained uninstantiated on the recorded DDR3 parity argument (scoria TASK-003).

The four verification items named above remain the coverage plan for the DDR3-specific behaviour; this edition does not claim they are all complete. What separates this edition from a 1.0 is stated below and filed as a task.

2026-10-03 (v0.9): TASK-001's bounded tranche exists behind CSRs and is now specified in this book — Modes A/B (elastic refresh, TCR) in Chapter 3.4, Mode C (ZQCS placement) in Chapter 3.2 — per the landed RTL, with per-mode design detail in `design-requirements.md` §6. Evidence (scoria TASK-001 close): 211 FUB + 21 macro + 4 top tests, Verilator and Verible lint green, and the refresh retention/accounting properties re-proved with the modes present in `formal/scoria/refresh_ctrl`. Block-by-block confirmation of the remaining INHERITED blocks is still TASK-008's pass.

17.4 Question resolutions (v0.2)

Five questions were listed in v0.1. Two are answered, one is struck as malformed, and two are deferred with a named condition.

Table 6.1: Question resolutions

#	Question	Resolution
Q1	DFI low-power handshake ordering on exit	**ANSWERED: the question does not arise on a Series-7 PHY.** See below
Q2	May periodic `ZQCS` preempt queued demand traffic?	**ANSWERED: no. Request/grant, sharing the maintenance sequencer**
Q3	Sample one prime DQ bit or several?	**ANSWERED: the prime bit only**, which makes `tWLOE` inert
Q4	The value of `tWLMRD`'s controller-defined maximum	**ANSWERED in kind: no PHY supplies one.** It is a policy value, not a measurement. See below
Q5	`REFpb` ordering: sequential or occupancy-aware?	**STRUCK — malformed.** The device owns the sequence (Ch 3.4)

17.4.1 Q1 and Q4: I was wrong to call these PHY-blocked

Both were deferred in v0.2 and v0.3 "pending a PHY choice". Examining the PHY shows that framing was wrong in both cases, and the correction matters because it unblocks the last two questions in this document.

The PHY choice is narrower than it looked. LiteDRAM's Series-7 PHY is one module, `s7ddrphy`, whose `A7DDRPHY` and `K7DDRPHY` classes differ by target family, not by DFI behaviour, and it supports DDR2 and DDR3 with write leveling. For any Xilinx 7-series target – which is every board this repository owns – the PHY is that module. There was no branch to wait on.

Q4: no PHY supplies a `tWLMRD` maximum, because none can. `s7ddrphy` exposes exactly two leveling controls, `_wlevel_en` (a CSR storage field) and `_wlevel_strobe` (a CSR pulse), and contains **no timeout of any kind**. JESD79-3F declares the maximum controller-dependent and the

PHY declines it too, so the value is a *policy* decision — how long to wait before declaring a leveling attempt failed — not a number to be measured off a PHY. scoria defines it, as this document already specified: a CSR timeout with its own status bit. Choosing the number is a one-line decision, not an experiment.

Q1: the question does not arise on this PHY. `s7ddrphy` implements **no DFI low-power interface at all** — no `lp_req`, no `lp_ack`, nothing. Nor does LiteDRAM's generated DDR3 controller: a search of the generated core for `dfi_lp` returns zero hits, while `CKE` appears 92 times. Power-down on this PHY family is achieved by **DRAM command** — CKE plus SRE/SRX — and not through the DFI low-power channel.

So there is no exit-ordering handshake to get wrong, because there is no counterparty to handshake with. The requirement changes accordingly:

- `powerdown_ctrl` achieves precharge-power-down, active-power-down and self-refresh through CKE and SRE/SRX, which is what pumice already does.
- scoria still *exposes* `dfi_lp_ctrl_req` and `dfi_lp_data_req`, because DFI v3.1 defines them and a future PHY may consume them. Their specified behaviour when no acknowledgement ever arrives is to **time out and report**, never to block power-down — an unacknowledged request must not be able to wedge the controller.
- The ordering question returns only with a PHY that implements the interface, and at that point it is answerable by reading that PHY rather than by speculation.

Note on the correction. This is the second framing error in this document's short life: v0.1 asked Q5, which the DRAM standard had already settled, and v0.2/v0.3 deferred Q1 and Q4 to a decision that turned out not to exist. Both came from the same habit — treating something as an open choice without first checking whether the thing it depends on had already answered it. The cheap check is to read the artifact.

17.4.2 Q2: ZQCS is a request, never a preemption

Answered from a DDR3 LiteDRAM core, generated for this purpose. v0.2 answered this by analogy, because the LiteDRAM core in the pumice flow is a DDR2 configuration and contains no ZQ logic at all. Sean's suggestion was to generate a DDR3 one; that core now exists and the analogy is no longer needed.

What the DDR3 core does (`bin/litedram_ddr3_ref.yml` regenerates it):

- **ZQCS shares the refresher's FSM.** There is one `refresher` state machine, and it starts the ZQCS executer from inside itself and waits for its done. Maintenance is not a separate agent competing for the bus; it is one sequencer issuing both kinds of maintenance command.

- **It requests, and waits.** `refresh_req` to every bank machine is driven from the maintenance command's valid. The bank machines must grant – drain and precharge – before the command issues. Nothing is preempted.
- **The executer is a timed sequence**, a trigger counter stepping to 19 and then asserting done, with the periodic timer held off (`zqcs_timer_wait`) until it completes. So `tZQCS` is enforced by holding the *next* interval, not by blocking traffic.

That is three independent controllers agreeing – LiteDRAM DDR2 refresh, LiteDRAM DDR3 ZQCS, and pumice refresh – so the baseline is settled rather than merely plausible: `scoria_zq_ctrl` raises a demand and waits for a grant.

Design consequence worth taking from LiteDRAM: it shares one sequencer between refresh and ZQ rather than giving ZQ its own. scoria's Chapter 2.3 specifies `scoria_zq_ctrl` as a separate FUB, and that remains the plan – pumice's `refresh_ctrl` is already substantial and adding a second maintenance mode to it would couple two independently-verifiable things. But the *interface* both present to the scheduler should be identical, so that the arbiter sees one kind of maintenance demand with a source tag rather than two special cases.

Condition to revisit unchanged: if characterization shows `ZQCS` starved under sustained demand – the interval slipping materially past its CSR value – placement becomes TASK-001's scheduling question. The issued-calibration counter required in Chapter 3.2 is what makes that measurable rather than suspected.

17.4.3 Q3: sample the prime DQ bit only

The write-leveling interface samples the prime DQ bit and not the full byte. The consequence is explicit: **`tWLOE` is inert in this implementation.** It exists in JESD79-3F to allow mismatch *across* DQ bits, which only matters to a controller sampling several.

`tWLOE` remains a CSR field so that a later implementation can sample wider without a register-map change, and it is documented as unused rather than omitted — an absent register is a silent limitation, while a present one documented as inert is a decision a reader can find.

17.4.4 Q5: struck rather than deferred

v0.1 asked whether `REFpb` round-robin should be sequential or occupancy-aware. There is nothing to choose: the `REFpb` command carries no bank address, the sequence is fixed in the device, and the controller's rotor only mirrors the device's counter. JESD209-3C states the sequence is fixed sequential round-robin; pumice's implementation already cites JESD209-2 §6.6 for the same property in LPDDR2.

It is struck rather than deferred because deferring implies a decision still exists. This one never did, and the correction also moved `refresh_ctrl` from MODIFIED to INHERITED — see Chapter 3.4.

17.5 What would make this a 1.0

Three things, in order:

1. A value chosen for `tWLMRD`'s maximum – Q4 reduced this to a one-line policy decision, and with CK at 2.5 ns on the named target a generous bound is easy to pick.
2. This document reconciled against the RTL block by block — every INHERITED marking either confirmed or corrected. v0.8 reconciled what implementation and the first board work had settled (revision history, Ch 0); v0.9 completed the pass for the two blocks TASK-001 had just changed (refresh in Ch 3.4, ZQ in Ch 3.2); the block-by-block confirmation of the remaining inherited blocks is still to do. A marking that turns out to be wrong is a defect in this document, and the correction goes here rather than being dropped silently.
3. The exact CSR map, which comes from the RDL and cannot honestly precede it. The RDL now exists (`rtl/macro/scoria_csr.rdl`, with generated documentation and Python regmap); finalizing the map against it is part of the same pass.

Until then this is a 0.9: a specification that matches what has been built and verified, and explicit about what remains unconfirmed.